Supplementary file

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29 **Table S1: PRISMA Checklist**
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Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	p.1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	p.2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	p.4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	p.5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	p.6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	p.5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supplement S2
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	p.6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	p.6-7

Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	p.7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	p.7-8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	p.8-9
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	p.9
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	p.7-8
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	p.9-10
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	p.10
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Figure 1, 11
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Figure 2, p.13
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Table 2, Supplement S6
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Table 3, p. 21
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Figure 2, p.14, Supplement S4
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Supplement S8, S9

DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	p.22-23
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	p.24
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	p.22, p.24
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	p.26

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

S2: Search strategy

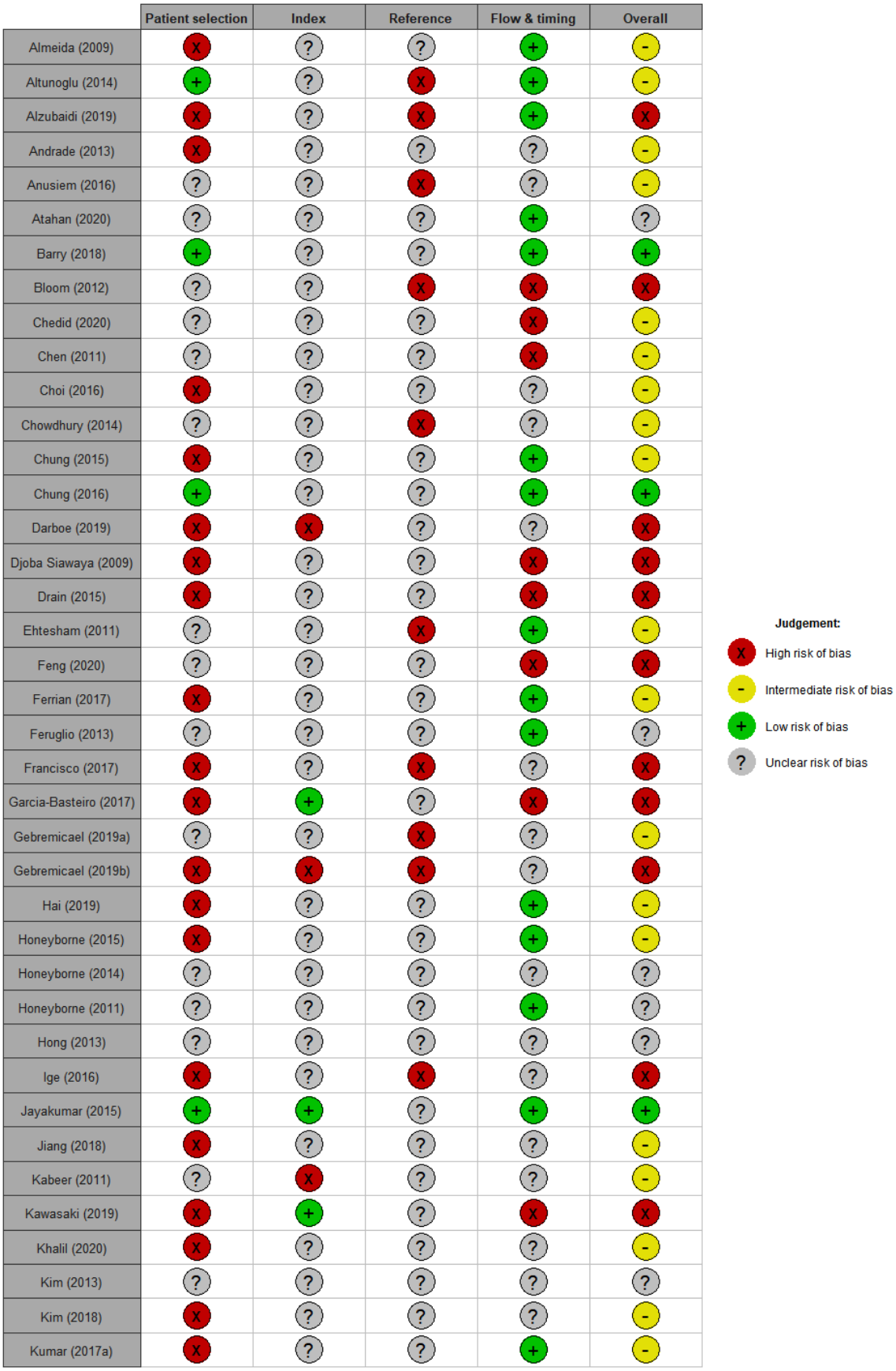
Search conducted in April 2019 with an updated search in October 2020:

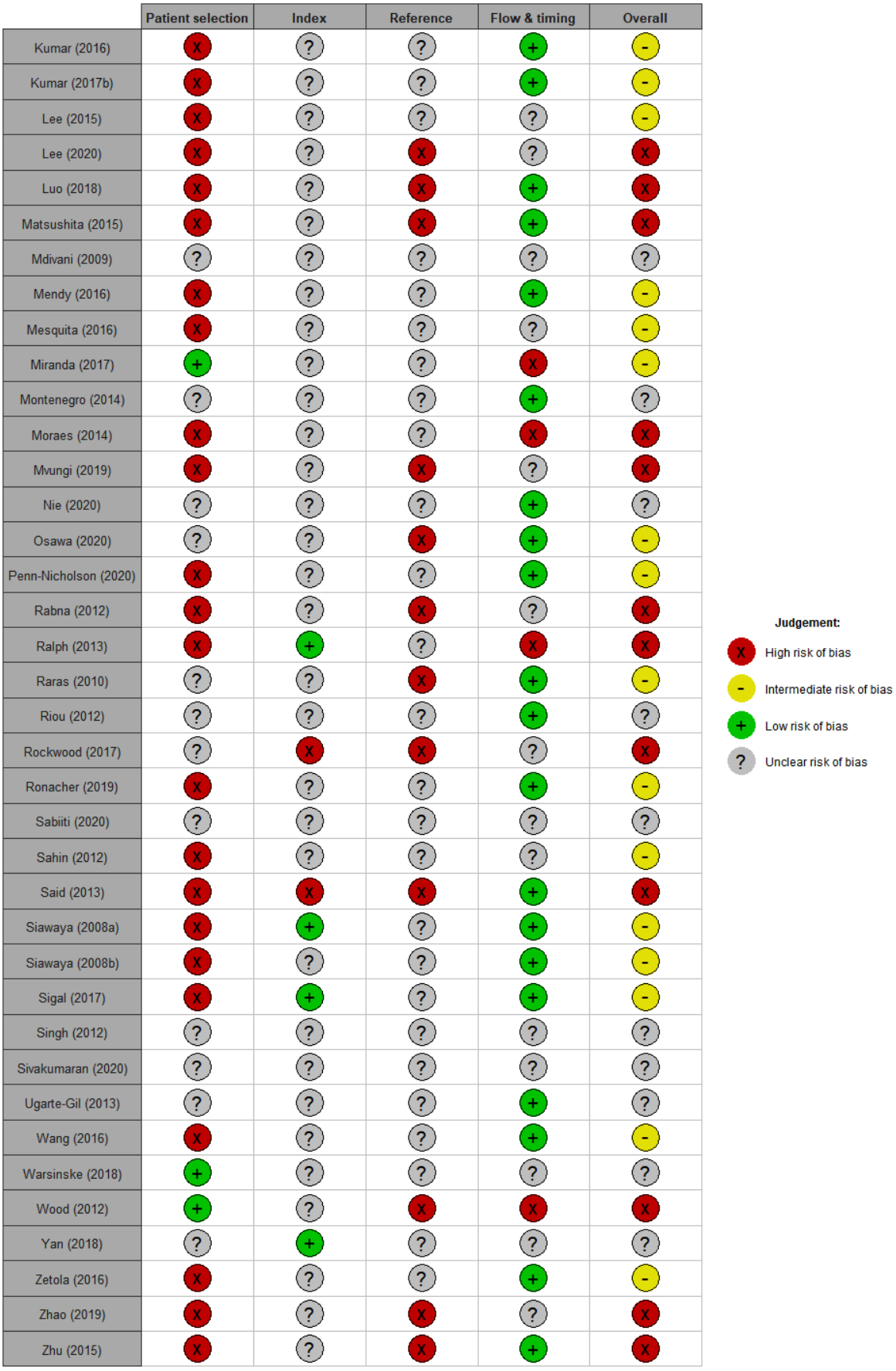
((((((("tuberculosis"[mh] OR "mycobacterium tuberculosis"[mh] OR "tuberculosis"[tiab] OR tb[tiab] OR MDRTB[tw] OR "Antitubercular Agents"[mh] OR antitubercul[tw])))) AND ((monitor*[tw] OR infectiousness[tiab] OR treatment outcome*[tiab] OR treatment response*[tiab] OR "treatment outcome"[mh] OR "microbial viability"[mh] OR viab*[tiab])))) AND (("biomarkers/blood"[mh] OR biomarker*[tw] OR immunologic test*[tw] OR eth-idium[tw] OR "propidium"[tw] OR "RNA"[tw] OR mRNA[tw] OR "DNA"[tw] OR lympho-cyte*[tw] OR Fluorescein Diacetate[tw] OR (t cell activation marker*[tw] AND assay*) OR gamma assay*[tw])))))*

Table S3: QUADS-2 guiding questions

Domain	Guiding questions
Patient selection	• Was a consecutive or random sample of patients or specimens enrolled? • Was a case-control design avoided? • Did the study avoid inappropriate exclusions?
Reference standard	• Is the reference standard likely to correctly classify the target conditions? • Were the reference standard results interpreted without knowledge of the results of the index test?
Index test	• Were the index test results interpreted without knowledge of the results of the reference standard? • Was the specimen processed as per the manufacturer's protocol of the particular index test?
Flow and timing	• Was there an appropriate interval between the index tests and reference standard? • Did all patients receive at least one identical reference standard? • Were all patients included in the analysis?

Figure S4: Study-specific QUADAS-2 scores




Patient selection
Index
Reference
Flow & timing
Overall
Kumar (2016)
Kumar (2017b)
Lee (2015)
Lee (2020)
Luo (2018)
Matsushita (2015)
Mdivani (2009)
Mendy (2016)
Mesquita (2016)
Miranda (2017)
Montenegro (2014)
Moraes (2014)
Mvungi (2019)
Nie (2020)
Osawa (2020)
Penn-Nicholson (2020)
Rabna (2012)
Ralph (2013)
Raras (2010)
Riou (2012)
Rockwood (2017)
Ronacher (2019)
Sabiiti (2020)
Sahin (2012)
Said (2013)
Siawaya (2008a)
Siawaya (2008b)
Sigal (2017)
Singh (2012)
Sivakumaran (2020)
Ugarte-Gil (2013)
Wang (2016)
Warsinske (2018)
Wood (2012)
Yan (2018)
Zetola (2016)
Zhao (2019)
Zhu (2015)
Judgement:
High risk of bias
Intermediate risk of bias
Low risk of bias
Unclear risk of bias

58 **Table S5: Complete list of studies and biomarkers included in qualitative analysis**
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Biomarker	Author (year)	Assay name	Ref	County	Follow ups	Sample used (state)	Reference used
16S rRNA	Hai (2019)	RT-qPCR for 16 S rRNA MBLA on the LightCycler Multiplex RNA Virus Master (Roche)	(1)	Vietnam	0, W1, W2, W4	Sputum (frozen)	Liquid culture (MGIT) Smear microscopy (ZN) GeneXpert (NR)
	Honeyborne (2011)	RT-qPCR for 16S rRNA MBLA on the Rotor-Gene Q 5-plex HRM platform (Qiagen)	(2)	South Africa	0, D3, D7, D14, D56	Sputum (frozen)	Liquid culture (7H9)
	Honeyborne (2014)	16S rRNA MBLA	(3)	Tanzania	0, D2, D4, D5, D7, D10, D14	Sputum (NR)	Solid culture (7H11) Liquid culture (MGIT 960)
	Sabiiti (2020)	RT-qPCR for 16s rRNA MBLA	(4)	Tanzania, Mozambique, Malawi	0, W1, W2, W3, W4, W5, W6, W7, W8, W9, W10, W11, W12	Sputum (NR)	Solid culture (LJ) Liquid culture (MGIT)
	Yan (2018)	Simultaneous amplification and testing for TB (SAT-TB) for 16S rRNA using AmpSure assay (Shanghai Rendu Biotechnology)	(5)	China	0, W2, M2, M5, M6-8	Sputum (NR)	Liquid culture (MGIT 960) Smear microscopy (FM) CT scan
85B mRNA	Atahan (2020)	RT-qPCR for 85B mRNA on the LightCycler 480 II Real-Time PCR System (Roche)	(6)	Turkey	0, D15, D30	Sputum (frozen)	Solid culture (LJ) Liquid culture (MGIT 960) Microscopy (NR)
	Montenegro (2014)	RT-qPCR for 85B mRNA on the ABI PRISM 7500 thermal cycler (Applied Biosystems)	(7)	Brazil	0, D15, D30	Sputum (fresh)	Solid culture (LJ)
	Mdivani (2009)	RT-PCR for 85B mRNA on the ABI PRISM 7700 Sequence Detection System (Applied Biosystems)	(8)	Georgia	0, W2, M1, M2, M4	Sputum (fresh)	Solid culture (LJ)
	Singh (2012)	qRT-PCR for 85B mRNA on the ABI Prism 7000 Sequence Detector System (Perkin Elmer Corp. / Applied Biosystems)	(9)	India	0, D30, D60	NR (NR)	Solid culture (LJ) Liquid culture (BACTEC 460) Smear microscopy (NR)
Adenosine deaminase (ADA)	Altunoglu (2014)	Serum ADA-colorimetric assay (Diazyme General Atomics)	(10)	Turkey	0, M2	Serum (frozen)	Smear microscopy (NR)
	Ige (2016)	ADA ELISA kit (Elabscience)	(11)	Nigeria	0, M6	Plasma (NR)	Smear microscopy (NR)

Albumin	De Moraes (2014)	Albumin Advia colorimetric quantification (Siemens Healthcare Diagnostics)	(12)	Brazil	0, D30, D60	Serum (fresh)	Solid culture (LJ) Clinical outcome Smear microscopy (ZN)
Angiostensinogen	Jiang (2018)	Angiotensinogen ELISA kit (RayBiotech)	(13)	China	0, M2	Serum (NR)	Culture (NR) Smear microscopy (NR)
Angiopoietins (Ang)	Kumar (2017)	Ang-1 and Ang-2 Quantikine ELISA KIT (R&D Systems)	(14)	India	0, M6	Plasma (frozen)	Solid culture (LJ)
Apolipoprotein	Jiang (2018)	Apolipoprotein C-II ELISA kit (RayBiotech)	(13)	China	0, M2	Serum (NR)	Culture (NR) Smear microscopy (NR)
	Jiang (2018)	Apolipoprotein A-IV ELISA kit (CUSABIO Biotech)	(13)	China	0, M2	Serum (NR)	Culture (NR) Smear microscopy (NR)
Beta-2 microglobulin (B2M)	Mendy (2016)	B2M ELISA kit (Immunology Consultants Laboratory)	(15)	Gambia	0, M2, M6	Plasma (frozen)	Liquid culture (MGIT) Smear microscopy (ZN) Chest X-ray
Cancer antigen 125 (CA-125)	Kim (2013)	CA-125 radio-immunoassay kit (Cis Bio International)	(16)	South Korea	0, M6	Serum (NR)	Culture (NR) Smear microscopy (NR) Chest X-ray
	Sahin (2012)	CA-125 microparticle enzyme immunoassay method (Abbott-Axsym System)	(17)	Turkey	0, M2	Serum (NR)	Solid culture (LJ) Smear microscopy (ZN) Chest X-ray
	Said (2013)	VIDAS CA-125 II (Biomerieux)	(18)	Egypt	0, M4	Serum (frozen)	Smear microscopy (ZN)
Complement C7	Jiang (2018)	Complement component C7 ELISA kit (Abcam)	(13)	China	0, M2	Serum (NR)	Culture (NR) Smear microscopy (NR)
Cough frequency	Lee (2020)	CayCaMo cough monitor	(19)	Peru	0, D3, D7, D14, D30, D60	Cough (fresh)	Smear microscopy (NR) Microscopic-observation drug susceptibility (MODS)
C-reactive protein (CRP)	Almeida (2009)	Roche CRPLX kit on the Roche Modular Analyser (Roche)	(20)	Brazil	0, W1, W3, W5, W8	Plasma (frozen)	Solid culture (NR) Clinical outcome Smear microscopy (NR)
	Djoba Siawaya (2008)	CRP ELISA kit (Bender MedSystems)	(21)	South Africa	0, W1, W5, W13, W26	Serum (frozen)	Chest X-ray Liquid culture (BACTEC)
	Ferrian (2017)	CRP on a multiplex (ProcartaPlex Human kits, eBioscience) read on a luminometer (Bioplex 200, BioRad)	(22)	South Africa	0, M2, M4, M6	Plasma (frozen)	Smear microscopy (NR) Liquid culture (MGIT 960) Line probe assay (Hain)

	Francisco (2017	Quantikine ELISA (R&D Systems Inc., Minneapolis, MN, USA)	(23)	China	0, M4, M6, M7	Whole blood (NR)	Clinical outcome
	Jayakumar (2015)	CRP iCHROMA Point-of-care Reader (BodiTech Med Inc)	(24)	Uganda	0, W8, W20	Serum (frozen)	Solid culture (NR) Liquid culture (MGIT 960)
	Khalil (2020)	CRP NycoCARD Reader II (Abbott)	(25)	Egypt	M1, M2, M3	Whole blood (NR) Plasma (NR) Serum (NR)	Solid culture (LJ) Clinical outcome Smear microscopy (ZN) GeneXpert (MTB/RIF)
	Mendy (2016)	CRP ELISA kit (Immunology Consultants Laboratory)	(15)	Gambia	0, M2, M6	Plasma (frozen)	Liquid culture (MGIT) Smear microscopy (ZN) Chest X-ray
	Mesquita (2016)	CRP ELISA kit (Ebioscience)	(26)	Brazil	0, D60	Serum (frozen)	Culture (NR) Chest X-ray
	Miranda (2017)	CRP ELISA kit (Ebioscience)	(27)	Brazil	0, D30, D60	Serum (frozen)	Solid culture (LJ)
	De Moraes (2014)	CRP BNII nephelometer (Dade Behring)	(12)	Brazil	0, D30, D60	Serum (fresh)	Solid culture (LJ) Clinical outcome Smear microscopy (ZN)
	Sigal (2017)	CRP – V-PLEX Human Vascular Injury Panel 2 (Meso Scale Diagnostics)	(28)	North America, Spain, South Africa, Uganda	0, W8, W12	Serum (frozen)	Solid culture (LJ) Liquid culture (MGIT 960) Clinical outcome Chest X-ray
CXCL9	Chung (2015)	CXCL9 ELISA kit (R&D Systems)	(29)	South Korea	0, M1, M2, M3, M6, M12	Serum (frozen)	Culture (NR) Smear microscopy (NR) Clinical outcome Chest X-ray
	Chung (2016)	CXCL9 ELISA kit (R&D Systems)	(30)	South Korea	0, M2	Serum (frozen)	Culture (NR) Smear microscopy (NR) Clinical outcome Chest X-ray
	Lee (2015)	CXCL9 – Bioplex Multiplex Suspension Array System (Bio-Rad Laboratories)	(31)	Taiwan	0, M2	Plasma (NR) Culture supernatant (NR)	Culture (NR)
CXCL11	Chung (2015)	CXCL11 ELISA kit (R&D Systems)	(29)	South Korea	0, M1, M2, M3, M6, M12	Serum (frozen)	Culture (NR) Smear microscopy (NR) Clinical outcome Chest X-ray

	Chung (2016)	CXCL11 ELISA kit (R&D Systems)	(30)	South Korea	0, M2	Serum (frozen)	Culture (NR) Smear microscopy (NR) Clinical outcome Chest X-ray
Endothelin-1 (ET-1)	Wang (2016)	ET-1 ELISA kit (R&D Systems)	(32)	China	0, D1, D2, D4, D6, D10, D14	Sputum (fresh)	Solid culture (7H11) Liquid culture (MGIT 960)
Eotaxin	Choi (2016)	Eotaxin – Millipore Human Cytokine/ Chemokine Magnetic Bead Panel (HCYTOMAG-60 K-02)	(33)	South Korea	0, M2, M6	Serum (fresh)	Solid culture (NR) Liquid culture (NR) Smear microscopy (NR)
	Choi (2016)	CCL11/Eotaxin Quantikine ELISA kit (R&D Systems Inc)	(33)	South Korea	0, M2, M6	Serum (fresh)	Solid culture (NR) Liquid culture (NR) Smear microscopy (NR)
	Djoba Siawaya (2009)	Eotaxin – Lincoplex human cytokine 29-plex assays (Millipore)	(34)	South Africa	0, W1, W5, W13, W26	Plasma (frozen)	Liquid culture (BACTEC 460T) Smear microscopy (NR)
	Riou (2012)	Exotaxin – Multiplex bead array (Bulletin# 10014905 Bio-Rad) on a luminometer (Luminex)	(35)	South Africa	0, W2, W4, W8, W12, W26, W52, W78	Plasma (frozen)	Solid culture (LJ) Liquid culture (MGIT) Smear microscopy (FM)
Epidermal growth factor (EGF)	Djoba Siawaya (2009)	EGF – Lincoplex human cytokine 29-plex assays (Millipore)	(34)	South Africa	0, W1, W5, W13, W26	Plasma (frozen)	Liquid culture (BACTEC 460T) Smear microscopy (NR)
Fractional exhaled nitric oxide (FeNO) & volatile organic compounds (VOCs)	Ralph (2013)	FeNO – Portable NiOX MINO device (Aerocrine, Sweden)	(36)	Indonesia	0, W1, W2, M1, M2, M6	Breath (fresh)	Liquid culture (MGIT) Smear microscopy (NR) Chest X-ray
	Zetola (2016)	VOC – quartz microbalance (QMB) gas sensors	(37)	Botswana	0, D2, D7, D14, D30	Breath (fresh)	Liquid culture (MGIT)
Fibroblast growth factor (FGF) basic	Riou (2012)	FGF basic – Multiplex bead array (Bulletin# 10014905 Bio-Rad) on a luminometer (Luminex)	(35)	South Africa	0, W2, W4, W8, W12, W26, W52, W78	Plasma (frozen)	Solid culture (LJ) Liquid culture (MGIT) Smear microscopy (FM)
Ferritin	Miranda (2017)	Ferritin ELISA kit (Abnova)	(27)	Brazil	0, D30, D60	Serum (frozen)	Solid culture (LJ)
Globulin	De Moraes (2014)	Globulin Advia colorimetric quantification (Siemens Healthcare Diagnostics)	(12)	Brazil	0, D30, D60	Serum (fresh)	Solid culture (LJ) Clinical outcome Smear microscopy (ZN)
Granulocyte colony stimulating factor (G-CSF)	Riou (2012)	G-CSF – Multiplex bead array (Bulletin# 10014905 Bio-Rad) on a luminometer (Luminex)	(35)	South Africa	0, W2, W4, W8, W12, W26, W52, W78	Plasma (frozen)	Solid culture (LJ) Liquid culture (MGIT) Smear microscopy (FM)

Granulocyte macrophage colony stimulating factor (GM-CSF)	Riou (2012)	GM-CSF – Multiplex bead array (Bulletin# 10014905 Bio-Rad) on a luminometer (Luminex))	(35)	South Africa	0, W2, W4, W8, W12, W26, W52, W78	Plasma (frozen)	Solid culture (LJ) Liquid culture (MGIT) Smear microscopy (FM)
Granzyme (A and B)	Djoba Siawaya (2008)	Granzyme B ELISA (Bender MedSystems)	(21)	South Africa	0, W1, W5, W13, W26	Serum (frozen)	Chest X-ray Liquid culture (BACTEC)
	Jayakumar (2015)	Granzyme B ELISA kit (Bender MedSystems)	(24)	Uganda	0, W8, W20	Serum (frozen)	Solid culture (NR) Liquid culture (MGIT 960)
	Lee (2015)	Granzyme A – Bioplex Multiplex Suspension Array System (Bio-Rad Laboratories)	(31)	Taiwan	0, M2	Plasma (NR) Culture supernatant (NR)	Culture (NR)
Heme oxygenase-1 (HO-1)	Andrade (2013)	HO-1 ELISA kit (Assay Designs)	(38)	India	W0, W26	Plasma (NR)	Culture (NR)
	Rockwood (2017)	HO-1 ELISA kit (ADI-EKS-800, Enzo Life Sciences)	(39)	South Africa	0, W8, W20	Plasma (frozen)	Culture (NR) Chest X-ray
Hemoglobin	De Moraes (2014)	Hemoglobin ABX Pentra DF 120 (Horiba Diagnostics)	(12)	Brazil	0, D30, D60	Serum (fresh)	Solid culture (LJ) Clinical outcome Smear microscopy (ZN)
High-sensitivity CRP (hs-CRP)	Altunoglu (2014)	hs-CRP BN II nephelometer (Dade Behring)	(10)	Turkey	0, M2	Serum (frozen)	Smear microscopy (NR)
	Chen (2011)	hs-CRP nephelometry (Dade Behring BM2)	(40)	Taiwan	0, M2, M6	Serum (NR)	Solid culture (LJ) Liquid culture (MGIT) Smear microscopy (ZN) Chest X-ray
	Ehtesham (2011)	hs-CRP ELISA kit (Diagnostic Biochem Canada Inc.)	(41)	India	0, M2, M4, M6	Serum (frozen)	Clinical outcome Smear microscopy (NR)
IFN-γ	Chowdhury (2014)	IFN-γ high sensitivity human ELISA kit (ImmunoTools)	(42)	India	0, M2, M4, M6	Serum (NR)	Smear microscopy (ZN) Chest X-ray
	Chung (2015)	IFN-y ELISA kit (R&D Systems)	(29)	South Korea	0, M1, M2, M3, M6, M12	Serum (frozen)	Culture (NR) Smear microscopy (NR) Clinical outcome Chest X-ray
	Djoba Siawaya (2008)	IFN-γ – Bio-Plex bead array system (Bio-Rad Laboratories)	(43)	South Africa	0, D3, W1, W2, W4, W8, W13, W26	Peripheral blood (frozen)	Smear microscopy (ZN) Liquid culture (BACTEC 460TB)
	Nie (2020)	IFN-γ ELISA kit (BioLegend)	(44)	China	0, M1-2, M6	Serum (frozen)	Culture (NR) Smear microscopy (NR) CT scan

	Matsushita (2015)	IFN-γ – 27-plex assay on the Bio-Plex Suspension Array System (Bio-Rad)	(45)	Vietnam	0, M2, M7	Plasma (frozen)	Smear microscopy (NR) Chest X-ray
	Mvungi (2019)	IFN-γ – multiplex assay (Human Premixed Multi-Analyte Kit; cat. # LXSAHM) on the Luminex 200 system	(46)	Tanzania	0, M2	Plasma (frozen) prepared using QuantiFERON-TB Gold Plus	Clinical outcome
	Riou (2012)	IFN-γ – Multiplex bead array (Bulletin# 10014905 Bio-Rad) on a luminometer (Luminex)	(35)	South Africa	0, W2, W4, W8, W12, W26, W52, W78	Plasma (frozen)	Solid culture (LJ) Liquid culture (MGIT) Smear microscopy (FM)
	Sigal (2017)	IFN-γ – V-PLEX Human Pro-inflammatory Panel 1 (Meso Scale Diagnostics)	(28)	North America, Spain, South Africa, Uganda	0, W8, W12	Serum (frozen)	Solid culture (LJ) Liquid culture (MGIT 960) Clinical outcome Chest X-ray
Iron	Gebremicael (2019)	Iron on c501 module of cobas6000 analyser (Roche Diagnostics)	(47)	Ethiopia	0, M6	Serum (frozen)	Clinical outcome Smear microscopy (ZN)
IS6110 insertion element	Mdivani (2009)	RT-DNA for IS6110 insertion using ABI PRISM 7700 Sequence Detection System (Applied Biosystems)	(8)	Georgia	0, W2, M1, M2, M4	Sputum (fresh)	Solid culture (LJ)
IL-1β	Anusiem (2016)	Interleukin-1β ELISA kit (eBioscience)	(48)	Nigeria	W0, W8, W26	Serum (frozen)	Smear microscopy (NR)
	Chowdhury (2014)	Human IL-1β ELISA kit (Raybiotech)	(42)	India	0, M2, M4, M6	Serum (NR)	Smear microscopy (ZN) Chest X-ray
	Luo (2018)	IL-1β ELISA kits (Siemens Healthcare Diagnostics)	(49)	China	0, M2	Serum (fresh)	Smear microscopy (FM)
	Riou (2012)	IL-1β – Multiplex bead array (Bulletin# 10014905 Bio-Rad) on a luminometer (Luminex)	(35)	South Africa	0, W2, W4, W8, W12, W26, W52, W78	Plasma (frozen)	Solid culture (LJ) Liquid culture (MGIT) Smear microscopy (FM)
IL-1RA	Matsushita (2015)	IL-1RA – 27-plex assay on the Bio-Plex Suspension Array System (Bio-Rad)	(45)	Vietnam	0, M2, M7	Plasma (frozen)	Smear microscopy (NR) Chest X-ray
	Riou (2012)	IL-1RA – Multiplex bead array (Bulletin# 10014905 Bio-Rad) on a luminometer (Luminex)	(35)	South Africa	0, W2, W4, W8, W12, W26, W52, W78	Plasma (frozen)	Solid culture (LJ) Liquid culture (MGIT) Smear microscopy (FM)

IL-2	Alzubaidi (2019)	IL-2 ELISA kit (R&D Systems)	(50)	Iraq	W0, D2, D4, W1, W2, W3, W4, W9	Sputum (fresh)	Smear microscopy (NR)
	Kumar (2017)	IL-2 Quantikine ELISA kit (R&D Systems)	47	India	0, M6	Plasma (fresh)	Culture (NR)
	Matsushita (2015)	IL-2 – 27-plex assay on the Bio-Plex Suspension Array System (Bio-Rad)	(45)	Vietnam	0, M2, M7	Plasma (frozen)	Smear microscopy (NR) Chest X-ray
	Riou (2012)	IL-2 – Multiplex bead array (Bulletin# 10014905 Bio-Rad) on a luminometer (Luminex)	(35)	South Africa	0, W2, W4, W8, W12, W26, W52, W78	Plasma (frozen)	Solid culture (LJ) Liquid culture (MGIT) Smear microscopy (FM)
IL-4	Chowdhury (2014)	IL-4 high sensitivity human ELISA kit (ImmunoTools)	(42)	India	0, M2, M4, M6	Serum (NR)	Smear microscopy (ZN) Chest X-ray
	Djoba Siawaya (2008)	IL4 – Bio-Plex bead array system (Bio-Rad Laboratories)	(43)	South Africa	0, D3, W1, W2, W4, W8, W13, W26	Peripheral blood (frozen)	Smear microscopy (ZN) Liquid culture (BACTEC 460TB)
	Mvungi (2019)	IL-4 multiplex assay (Human Premixed Multi-Analyte Kit; cat. # LXSAHM) on the Luminex 200 system	(46)	Tanzania	0, M2	Plasma (frozen)	Clinical outcome
	Nie (2020)	IL-4 ELISA kit (BioLegend, USA)	(44)	China	0, M1-2, M6	Serum (frozen)	Culture (NR) Smear microscopy (NR) Chest computed tomography
	Riou (2012)	IL-4 – Multiplex bead array (Bulletin# 10014905 Bio-Rad) on a luminometer (Luminex)	(35)	South Africa	0, W2, W4, W8, W12, W26, W52, W78	Plasma (frozen)	Solid culture (LJ) Liquid culture (MGIT) Smear microscopy (FM)
IL-5	Riou (2012)	IL-5 – Multiplex bead array (Bulletin# 10014905 Bio-Rad) on a luminometer (Luminex)	(35)	South Africa	0, W2, W4, W8, W12, W26, W52, W78	Plasma (frozen)	Solid culture (LJ) Liquid culture (MGIT) Smear microscopy (FM)
IL-6	Chowdhury (2014)	IL-6 ELISA kit (Raybiotech)	(42)	India	0, M2, M4, M6	Serum (NR)	Smear microscopy (ZN) Chest X-ray
	Djoba Siawaya (2009)	IL-6 – Lincoplex human cytokine 29-plex assays (Millipore)	(34)	South Africa	0, W1, W5, W13, W26	Plasma (frozen)	Liquid culture (BACTEC 460T) Smear microscopy (NR)
	Feng (2020)	IL-6 Quantikine ELISA kit (R&D Systems)	(51)	Taiwan	0, W8	Peripheral blood mononuclear cells (frozen)	Culture (NR) Smear microscopy (NR)

	Luo (2018)	IL-6 ELISA kits (Siemens Healthcare Diagnostics Products Ltd., Llanberis, Gwynedd, UK)	(49)	China	0, M2	Serum (fresh)	Smear microscopy (FM)
	Mvungi (2019)	IL-6 multiplex assay (Human Premixed Multi-Analyte Kit; cat. # LXSAHM) on the Luminex 200 system	(46)	Tanzania	0, M2	Plasma (frozen) prepared using QuantiFERON-TB Gold Plus	Clinical outcome
	Riou (2012)	IL-6 – Multiplex bead array (Bulletin# 10014905 Bio-Rad) on a luminometer (Luminex)	(35)	South Africa	0, W2, W4, W8, W12, W26, W52, W78	Plasma (frozen)	Solid culture (LJ) Liquid culture (MGIT) Smear microscopy (FM)
	Sigal (2017)	IL-6 – V-PLEX Human Pro-inflammatory Panel 1 (Meso Scale Diagnostics)	(28)	North America, Spain, South Africa, Uganda	0, W8, W12	Serum (frozen)	Solid culture (LJ) Liquid culture (MGIT 960) Clinical outcome Chest X-ray
IL-7	Kumar (2017)	IL-7 Quantikine ELISA kit (R&D Systems)	(52)	India	0, M6	Plasma (fresh)	Culture (NR)
	Riou (2012)	IL-7 – Multiplex bead array (Bulletin# 10014905 Bio-Rad) on a luminometer (Luminex)	(35)	South Africa	0, W2, W4, W8, W12, W26, W52, W78	Plasma (frozen)	Solid culture (LJ) Liquid culture (MGIT) Smear microscopy (FM)
IL-9	Riou (2012)	IL-9 – Multiplex bead array (Bulletin# 10014905 Bio-Rad) on a luminometer (Luminex)	(35)	South Africa	0, W2, W4, W8, W12, W26, W52, W78	Plasma (frozen)	Solid culture (LJ) Liquid culture (MGIT) Smear microscopy (FM)
IL-10, IL-10Ra	Chowdhury (2014)	IL-10 high sensitivity human ELISA set (ImmunoTools)	(42)	India	0, M2, M4, M6	Serum (NR)	Smear microscopy (ZN) Chest X-ray
	Djoba Siawaya (2009)	IL-10 – Lincoplex human cytokine 29-plex assays (Millipore)	(34)	South Africa	0, W1, W5, W13, W26	Plasma (frozen)	Liquid culture (BACTEC 460T) Smear microscopy (NR)
	Mvungi (2019)	IL-10 – multiplex assay (Human Premixed Multi-Analyte Kit; cat. # LXSAHM) on the Luminex 200 system	(46)	Tanzania	0, M2	Plasma (frozen) prepared using QuantiFERON-TB Gold Plus	Clinical outcome
	Riou (2012)	IL-10 – Multiplex bead array (Bulletin# 10014905 Bio-Rad) on a luminometer (Luminex)	(35)	South Africa	0, W2, W4, W8, W12, W26, W52, W78	Plasma (frozen)	Solid culture (LJ) Liquid culture (MGIT) Smear microscopy (FM)
	Zhao (2019)	IL-10Ra ELISA kit (accession #: Q13651; RayBiotech)	(53)	China	0, W1, W2, W3, W4, W5, W6, W7, W8	Peripheral blood mononuclear cells (fresh)	Smear microscopy (NR)

IL-12, IL-12p40, IL-12p700	Djoba Siawaya (2009)	IL-12p40 – Lincoplex human cytokine 29-plex assays (Millipore)	(34)	South Africa	0, W1, W5, W13, W26	Plasma (frozen)	Liquid culture (BACTEC 460T) Smear microscopy (NR)
	Chowdhury (2014)	IL-12p40 ELISA Kit (RayBiotech)	(42)	India	0, M2, M4, M6	Serum (NR)	Smear microscopy (ZN) Chest X-ray
	Riou (2012)	IL-12p70 – Multiplex bead array (Bulletin# 10014905 Bio-Rad) on a luminometer (Luminex))	(35)	South Africa	0, W2, W4, W8, W12, W26, W52, W78	Plasma (frozen)	Solid culture (LJ) Liquid culture (MGIT) Smear microscopy (FM)
	Zhu (2015)	IL-12 – ELISA kit (eBioscience)	(54)	China	0, W2-8	Plasma (frozen)	Smear microscopy (NR)
IL-13	Djoba Siawaya (2009)	IL-13 – Lincoplex human cytokine 29-plex assays (Millipore)	(34)	South Africa	0, W1, W5, W13, W26	Plasma (frozen)	Liquid culture (BACTEC 460T)
	Riou (2012)	IL-13 – Multiplex bead array (Bulletin# 10014905 Bio-Rad) on a luminometer (Luminex)	(35)	South Africa	0, W2, W4, W8, W12, W26, W52, W78	Plasma (frozen)	Solid culture (LJ) Liquid culture (MGIT) Smear microscopy (FM)
IL-15	Djoba Siawaya (2009)	IL-15 – Lincoplex human cytokine 29-plex assays (Millipore)	(34)	South Africa	0, W1, W5, W13, W26	Plasma (frozen)	Liquid culture (BACTEC 460T) Smear microscopy (NR)
	Kumar (2017)	IL-15 Quantikine ELISA kit (R&D Systems)	(52)	India	0, M6	Plasma (fresh)	Culture (NR)
	Riou (2012)	IL-15 – Multiplex bead array (Bulletin# 10014905 Bio-Rad) on a luminometer (Luminex)	(35)	South Africa	0, W2, W4, W8, W12, W26, W52, W78	Plasma (frozen)	Solid culture (LJ) Liquid culture (MGIT) Smear microscopy (FM)
IL-17	Chen (2011)	IL-17 ELISA (R&D Systems)	(40)	Taiwan	0, M2, M6	Serum (NR)	Solid culture (LJ) Liquid culture (MGIT) Smear microscopy (ZN) Chest X-ray
	Riou (2012)	IL-17 – Multiplex bead array (Bulletin# 10014905 Bio-Rad) on a luminometer (Luminex)	(35)	South Africa	0, W2, W4, W8, W12, W26, W52, W78	Plasma (frozen)	Solid culture (LJ) Liquid culture (MGIT) Smear microscopy (FM)
IL-17A	Feng (2020)	IL-17A – 7-Plex Human ProcartaPlex Panel (Affymetrix eBioscience)	(51)	Taiwan	0, W8	Peripheral blood (frozen)	Culture (NR) Smear microscopy (NR)
IL-18	Lee (2015)	IL-18 – Bioplex Multiplex Suspension Array System (Bio-Rad Laboratories)	(31)	Taiwan	0, M2	Plasma (NR) Culture supernatant (NR)	Culture (NR)

IL-21	Feng (2020)	IL-21 – 7-Plex Human ProcartaPlex Panel (Affymetrix eBioscience)	(51)	Taiwan	0, W8	Peripheral blood (frozen)	Culture (NR) Smear microscopy (NR)
	Kumar (2017)	IL-21 Quantikine ELISA kit (R&D Systems)	(52)	India	0, M6	Plasma (fresh)	Culture (NR)
IL-22	Feng (2020)	IL-22 – 7-Plex Human ProcartaPlex Panel (Affymetrix eBioscience)	(51)	Taiwan	0, W8	Peripheral blood (frozen)	Culture (NR) Smear microscopy (NR)
	Lee (2015)	IL-22 – Bioplex Multiplex Suspension Array System (Bio-Rad Laboratories)	(31)	Taiwan	0, M2	Plasma (NR) Culture supernatant (NR)	Culture (NR)
IL-23	Feng (2020)	IL-23 – 7-Plex Human ProcartaPlex Panel (Affymetrix eBioscience)	(51)	Taiwan	0, W8	Peripheral blood (frozen)	Culture (NR) Smear microscopy (NR)
IP-10 (CXCL10)	Chen (2011)	IP-10 ELISA (R&D Systems)	(40)	Taiwan	0, M2, M6	Serum (NR)	Solid culture (LJ) Liquid culture (MGIT) Smear microscopy (ZN) Chest X-ray
	Chung (2016)	IP-10 ELISA (R&D systems, Minneapolis, MN)	(30)	South Korea	0, M2	Serum (frozen)	Culture (NR) Smear microscopy (NR) Clinical outcome Chest X-ray
	Djoba Siawaya (2009)	IP-10 – Lincoplex human cytokine 29-plex assays (Millipore)	(34)	South Africa	0, W1, W5, W13, W26	Plasma (frozen)	Liquid culture (BACTEC 460T) Smear microscopy (NR)
	Ferrian (2017)	IP-10 on a multiplex (ProcartaPlex Human kits, eBioscience) read on a luminometer (Bioplex 200, BioRad)	(22)	South Africa	0, M2, M4, M6	Plasma (frozen)	Smear microscopy (NR) Liquid culture (MGIT 960)
	Francisco (2017)	ELISA kit (Raybiotech, Inc.)	(23)	China	0, M4, M6, M7	Whole blood (NR)	Clinical outcome
	Garcia-Basteiro (2017)	IP-10 ELISA kit (Becton Dickinson and Company)	(55)	Mozambique	0, D7, D60	Serum (frozen)	Smear microscopy (ZN) GeneXpert (MTB/RIF) Liquid culture (MGIT 960)
	Hong (2013)	IP-10 ELISA kit (R&D Systems)	(56)	South Korea	0/within W2, after M6-M9	Serum (NR)	Culture (NR) Chest X-ray CT scan

	Jayakumar (2015)	IP-10 ELISA kit (R&D Systems)	(24)	Uganda	0, W8, W20	Serum (frozen)	Solid culture (NR) Liquid culture (MGIT 960)
	Kabeer (2011)	IP-10 ELISA kit (R&D Sysytems) in response to QFT-IT and RD1	(57)	India	0, M6	Plasma (fresh)	Solid culture (LJ) Liquid culture (BacT)
	Kim (2018)	IP-10 ELISA kit (R&D Systems, Minneapolis, MN, USA)	(58)	South Korea	0, M6, M12	Urine (NR), Serum (NR)	Culture (NR) Chest X-ray
	Matsushita (2015)	IP-10 – 27-plex assay on the Bio-Plex Suspension Array System (Bio-Rad)	(45)	Vietnam	0, M2, M7	Plasma (frozen)	Smear microscopy (NR) Chest X-ray
	Riou (2012)	IP-10 – Multiplex bead array (Bulletin# 10014905 Bio-Rad) on a luminometer (Luminex)	(35)	South Africa	0, W2, W4, W8, W12, W26, W52, W78	Plasma (frozen)	Solid culture (LJ) Liquid culture (MGIT) Smear microscopy (FM)
	Zhu (2015)	IP-10 ELISA kit (eBioscience)	(54)	China	0, W2-8	Plasma (frozen)	Smear microscopy (NR)
KL-6	Djoba Siawaya (2009)	KL-6 ELISA kit (Sanko Junyaku)	(34)	South Africa	0, W1, W5, W13, W26	Plasma (frozen)	Liquid culture (BACTEC 460T)
Leukocyte immunoglobulin like receptor B4 (LILRB4)	Zhao (2019)	SLAMF8 ELISA kit (accession #: Q8NHJ6; RayBiotech)	(53)	China	0, W1, W2, W3, W4, W5, W6, W7, W8	Peripheral blood mononuclear cells (fresh)	Smear microscopy (NR)
Lipoarabinomannan (LAM)	Drain (2015)	Determine TB LAM test (Alere Inc.)	(59)	South Africa	0, M2, EOT	Urine (frozen)	Liquid culture (MGIT) Solid culture (NR) Clinical outcome
	Kawasaki (2019)	LAM ELISA kit (Otsuka Pharmaceuticals)	(60)	Philippines	0, D7, D14, D28, D56	Sputum (NR)	Liquid culture (MGIT) Smear microscopy (NR) TB-LAMP
	Wood (2012)	LAM ELISA kit (Alere Health Services)	(61)	South Africa	0, W1, W2, W8, W16, W24	Urine (frozen)	GeneXpert (MTB/RIF)
	Feruglio (2013)	Limulus amebocyte lysate colorimetric assay (Lonza) to detect TB-LAM from the LAM ELISA kit (Wuhan Eiaab Science)	(62)	Norway	0, W2, W8, W24	Plasma (frozen)	Culture (NR) Smear microscopy (NR)
Lipocalcin-2	Choi (2016)	Lipocalcin-2 – Millipore Human Cytokine/ Chemokine Magnetic Bead Panel (HCYTOMAG-60 K-02)	(33)	South Korea	0, M2, M6	Serum (fresh)	Solid culture (NR) Liquid culture (NR) Smear microscopy (NR)
	Choi (2016)	MIP-1α – Millipore Human Cytokine/ Chemokine Magnetic	(33)	South Korea	0, M2, M6	Serum (fresh)	Solid culture (NR) Liquid culture (NR)

Macrophage inflammatory proteins (MIP)		Bead Panel (HCYTOMAG-60 K-02)					Smear microscopy (NR)
	Djoba Siawaya (2009)	MIP-1α & MIP-1β – Lincoplex human cytokine 29-plex assays (Millipore)	(34)	South Africa	0, W1, W5, W13, W26	Plasma (frozen)	Liquid culture (BACTEC 460T) Smear microscopy (NR)
	Riou (2012)	MIP-1β – Multiplex bead array (Bulletin# 10014905 Bio-Rad) on a luminometer (Luminex)	(35)	South Africa	0, W2, W4, W8, W12, W26, W52, W78	Plasma (frozen)	Solid culture (LJ) Liquid culture (MGIT) Smear microscopy (FM)
Matrix metalloproteinase (MMP)	Lee (2015)	MMP-8 – Bioplex Multiplex Suspension Array System (Bio-Rad Laboratories)	(31)	Taiwan	0, M2	Plasma (NR) Culture supernatant (NR)	Culture (NR)
	Sigal (2017)	MMP-8 – TB Panel 4 custom designed (Meso Scale Diagnostics)	(28)	North America, Spain, South Africa, Uganda	0, W8, W12	Serum (frozen)	Solid culture (LJ) Liquid culture (MGIT 960) Clinical outcome Chest X-ray
	Ugarte-Gil (2013)	MMP-1, -2, -3, -7, -8, -9 using Luminex multiplex array (R&D Systems)	(63)	Peru	0, W2, W8, W24	Sputum (frozen)	Liquid culture (NR) Clinical outcome Smear microscopy (FM)
Monocyte chemotactic protein1 (MCP-1)	Djoba Siawaya (2009)	MCP-1 – Lincoplex human cytokine 29-plex assays (Millipore)	(34)	South Africa	0, W1, W5, W13, W26	Plasma (frozen)	Liquid culture (BACTEC 460T) Smear microscopy (NR)
	Ferrian (2017)	MCP-1 on a multiplex (ProcartaPlex Human kits, eBioscience) read on a luminometer (Bioplex 200, BioRad)	(22)	South Africa	0, M2, M4, M6	Plasma (frozen)	Smear microscopy (NR) Liquid culture (MGIT 960) Line probe assay (Hain)
	Riou (2012)	MCP-1 – Multiplex bead array (Bulletin# 10014905 Bio-Rad) on a luminometer (Luminex)	(35)	South Africa	0, W2, W4, W8, W12, W26, W52, W78	Plasma (frozen)	Solid culture (LJ) Liquid culture (MGIT) Smear microscopy (FM)
Neopterin	Altunoglu (2014)	Neopterin ELISA KIT (Immuno-Biological Laboratories)	(10)	Turkey	0, M2	Serum (frozen)	Smear microscopy (NR)
	Jayakumar (2015)	Neopterin ELISA kit (Immuno-Biological Laboratories)	(24)	Uganda	W0, W8, W20	Serum (frozen)	Solid culture (NR) Liquid culture (MGIT 960)
	Mendy (2016)	Neopterin ELISA kit (Immuno-Biological Laboratories)	(15)	Gambia	0, M2, M6	Plasma (frozen)	Liquid culture (MGIT) Smear microscopy (ZN) Chest X-ray
Osteopontin (OPN)	Zhu (2015)	OPN ELISA kit (R&D Systems)	(54)	China	0, W2-8	Plasma (frozen)	Smear microscopy (NR)

Pentraxin-3 (PTX-3)	Sigal (2017)	PTX-3 – TB Panel 2 custom designed (Meso Scale Diagnostics)	(28)	North America, Spain, South Africa, Uganda	0, W8, W12	Serum (frozen)	Solid culture (LJ) Liquid culture (MGIT 960) Clinical outcome Chest X-ray
Procalcitonin	Osawa (2020)	µTASWako i30 immunoanalyzer (Fujifilm Wako Pure Chemical)	(64)	South Africa	0, D7, D14, D28	Serum (NR)	Smear microscopy (NR) Chest X-ray
Platelet derived growth factor (PDGF) BB	Riou (2012)	PDGF-BB – Multiplex bead array (Bulletin# 10014905 Bio-Rad) on a luminometer (Luminex)	(35)	South Africa	0, W2, W4, W8, W12, W26, W52, W78	Plasma (frozen)	Solid culture (LJ) Liquid culture (MGIT) Smear microscopy (FM)
RANTES	Lee (2015)	RANTES – Bioplex Multiplex Suspension Array System (Bio-Rad Laboratories)	(31)	Taiwan	0, M2	Plasma (NR) Culture supernatant (NR)	Culture (NR)
Resistin	Ehtesham (2011)	Human resistin ELISA kit (AdipoGen)	(41)	India	0, M2, M4, M6	Serum (frozen)	Clinical outcome
sCD14	Feruglio (2013)	sCD14 ELISA kit (R&D Systems)	(62)	Norway	0, W2, W8, W24	Plasma (frozen)	Culture (NR) Smear microscopy (NR)
sCD40L	Djoba Siawaya (2009)	sCD40L – Lincoplex human cytokine 29-plex assays (Millipore)	(34)	South Africa	0, W1, W5, W13, W26	Plasma (frozen)	Liquid culture (BACTEC 460T) Smear microscopy (NR)
sICAM-1	Djoba Siawaya (2008)	sICAM-1 ELISA kit (R&D Systems)	(21)	South Africa	0, W1, W5, W13, W26	Serum (frozen)	Chest X-ray Liquid culture (BACTEC)
	Jayakumar (2015)	sICAM-1 ELISA kit (R&D Systems)	(24)	Uganda	0, W8, W20	Serum (frozen)	Solid culture (NR) Liquid culture (MGIT 960)
Serum amyloid A (SAA) and serum amyloid protein A (SAA1)	Ferrian (2017)	SAA on a multiplex (ProcartaPlex Human kits, eBioscience) read on a luminometer (Bioplex 200, BioRad)	(22)	South Africa	0, M2, M4, M6	Plasma (frozen)	Smear microscopy (NR) Liquid culture (MGIT 960) Line probe assay (Hain)
	Sigal (2017)	SAA1 – V-PLEX Human Vascular Injury Panel 2 (Meso Scale Diagnostics)	(28)	North America, Spain, South Africa, Uganda	0, W8, W12	Serum (frozen)	Solid culture (LJ) Liquid culture (MGIT 960) Clinical outcome Chest X-ray
SLAMF8	Zhao (2019)	SLAMF8 ELISA kit (accession #: Q9P0V8; RayBiotech)	(53)	China	0, W1, W2, W3, W4, W5, W6, W7, W8	Peripheral blood mononuclear cells (fresh)	Smear microscopy (NR)
Small RNA (smRNA) and microRNA (miRNA)	Barry (2018)	5 plasma miRNA 10 plasma miRNA	(65)	China	0, M1, M2, M6	Plasma (frozen)	Culture (NR) Smear microscopy (NR) Chest X-ray Clinical outcome

	Honeyborne (2015)	miR-29A, SNORD61, miR-17-3p, miR-133a using Rotorgene-100 human serum/plasma MIHS-106Z arrays (SABiosciences)	(66)	South Africa	0, W24	Plasma (frozen)	Culture (NR)
Soluble death receptor (sDR)	Djoba Siawaya (2008)	sDR5 – BioPlex bead array system (Bio-Rad Laboratories) on Luminex	(21)	South Africa	0, W1, W5, W13, W26	Serum (frozen)	Chest X-ray Liquid culture (BACTEC)
Soluble lymphocyte activation gene-3 (sLAG-3)	Djoba Siawaya (2008)	sLAG-3 ELISA kit (Apotech)	(21)	South Africa	0, W1, W5, W13, W26	Serum (frozen)	Chest X-ray Liquid culture (BACTEC)
Soluble receptor analytes	Choi (2016)	sIL-2Ra – Millipore Human Cytokine/ Chemokine Magnetic Bead Panel (HCYTOMAG-60 K-02)	(33)	South Korea	0, M2, M6	Serum (fresh)	Solid culture (NR) Liquid culture (NR) Smear microscopy (NR)
	Ferrian (2017)	sIL2Ra on a multiplex (ProcartaPlex Human kits, eBioscience) read on a luminometer (Bioplex 200, BioRad)	(22)	South Africa	0, M2, M4, M6	Plasma (frozen)	Smear microscopy (NR) Liquid culture (MGIT 960) Line probe assay (Hain)
	Jayakumar (2015)	sIL-2Ra, sTNF-R1, and sTNF-R2 on the Luminex 100 multiplex platform (Millipore Corporation)	(24)	Uganda	0, W8, W20	Serum (frozen)	Solid culture (NR) Liquid culture (MGIT 960)
	Luo (2018)	sIL-2R ELISA kit (Siemens Healthcare Diagnostics)	(49)	China	0, M2	Serum (fresh)	Smear microscopy (FM)
	Nie (2020)	sIL-2R ELISA kit (BioLegend)	(44)	China	0, M1-2, M6	Serum (frozen)	Culture (NR) Smear microscopy (NR) CT scan
Soluble tumor necrosis factor receptor (sTNFR)	Djoba Siawaya (2008)	sTNF I, sTNFR II – BioPlex bead array system (Bio-Rad Laboratories) on Luminex	(21)	South Africa	0, W1, W5, W13, W26	Serum (frozen)	Chest X-ray Liquid culture (BACTEC)
Soluble urokinase-type plasminogen activator receptor (suPAR)	Jayakumar (2015)	suPAR ELISA kit (ViroGates)	(24)	Uganda	0, W8, W20	Serum (frozen)	Solid culture (NR) Liquid culture (MGIT 960)
	Rabna (2012)	suPAR ELISA kit (ViroGates)	(67)	Guinea-Bissau	0, W2, M1, M2, M5, M8	Plasma (frozen)	Clinical outcome Smear microscopy (ZN)
	Raras (2010)	suPAR ELISA kit (ViroGates)	(68)	Indonesia	0, M2, M4, M6	Serum (frozen)	Clinical outcome Smear microscopy (NR) Chest X-ray

	Ronacher (2019)	suPAR ELISA kit (Virogates)	(69)	South Africa	0, W1, W2, W4, W6 W26	Serum (frozen)	Liquid culture (Bactec 460)
	Djoba Siawaya (2008)	suPAR ELISA kit (ViroGates)	(21)	South Africa	0, W1, W5, W13, W26	Serum (frozen)	Chest X-ray Liquid culture (BACTEC)
TGF-β1	Chowdhury (2014)	TGF-β1 ELISA Kit (RayBiotech)	(42)	India	0, M2, M4, M6	Serum (NR)	Smear microscopy (ZN) Chest X-ray
Tissue inhibitor metalloproteinase (TIMP)	Ugarte-Gil (2013)	TIMP-1 & TIMP-2 using ELISA kit (R&D Systems)	(63)	Peru	0, W2, W8, W24	Sputum (frozen)	Liquid culture (NR) Clinical outcome Smear microscopy (FM)
TNF-α	Chowdhury (2014)	TNF-α high sensitivity human ELISA set (ImmunoTools)	(42)	India	0, M2, M4, M6	Serum (NR)	Smear microscopy (ZN) Chest X-ray
	Djoba Siawaya (2009)	TNF-α – Lincoplex human cytokine 29-plex assays (Millipore)	(34)	South Africa	0, W1, W5, W13, W26	Plasma (frozen)	Liquid culture (BACTEC 460T) Smear microscopy (NR)
	Luo (2018)	TNF-α ELISA kit (Siemens Healthcare Diagnostics)	(49)	China	0, M2	Serum (fresh)	Smear microscopy (FM)
	Mvungi (2019)	TNF-α – multiplex assay (Human Premixed Multi-Analyte Kit; cat. # LXSAHM) on the Luminex 200 system	(46)	Tanzania	0, M2	Plasma (frozen) prepared using QuantiFERON-TB Gold Plus	Clinical outcome
	Nie (2020)	TNF-α ELISA kit (BioLegend)	(44)	China	0, M1-2, M6	Serum (frozen)	Culture (NR) Smear microscopy (NR) Chest computed tomography
	Riou (2012)	TNF-α – Multiplex bead array (Bulletin# 10014905 Bio-Rad) on a luminometer (Luminex))	(35)	South Africa	0, W2, W4, W8, W12, W26, W52, W78	Plasma (frozen)	Solid culture (LJ) Liquid culture (MGIT) Smear microscopy (FM)
	Zhu (2015)	TNF-α ELISA kit (eBioscience)	(54)	China	0, W2-8	Plasma (frozen)	Smear microscopy (NR)
Transcriptomic/ gene signatures	Bloom (2012)	664-transcript signature 320-transcript signature	(70)	South Africa	0, W2, M2, M6, M12	Whole blood (frozen)	Clinical outcome Chest X-ray
	Darboe (2019)	RISK11 signature	(71)	South Africa	0, M2, M6, M8,M14	Whole blood (NR)	Culture (NR)
	Francisco (2017)	3-gene signature (*GBP5*, *DUSP3* and *KLF2*)	(23)	China	0, M4, M6, M7	Whole blood (NR)	Clinical outcome
	Gebremicael (2019)	105 genes expression profiling by dual-color Reverse-Transcription Multiplex Ligation-dependent	(72)	Ethiopia	0, M6, M18	Whole blood (NR)	Smear microscopy (ZN)

		Probe Amplification (dcRT-MLPA) platform					
	Penn-Nicholson (2020)	RISK6 signature	(73)	South Africa	0, M2, treatment completion, 6-8 76months post-treatment	Whole blood (frozen)	Culture (NR) Smear microscopy (NR) GeneXpert (MTB/RIF)
	Sivakumaran (2020)	198 gene set profiled using dual-colour-Reverse-Transcriptase-Multiplex-Ligation-dependent-Probe-Amplification (dc-RT MLPA) platform	(74)	India	0, M1, M2, M6	Whole blood (frozen)	Liquid culture (MGIT) Smear microscopy (FM) Clinical outcome
	Warsinske (2018)	3-gene signature (*GBP5*, *DUSP3* and *KLF2*)	(75)	South Africa	0, W1, W4, W24	Whole blood (NR)	Liquid culture (MGIT) PET-CT
Vascular endothelial growth factor (VEGF)	Djoba Siawaya (2009)	VEGF – Lincoplex human cytokine 29-plex assays (Millipore)	(34)	South Africa	0, W1, W5, W13, W26	Plasma (frozen)	Liquid culture (BACTEC 460T) Smear microscopy (NR)
	Ferrian (2017)	VEGF-A on a multiplex (ProcartaPlex Human kits, eBioscience) read on a Bioplex 200 (BioRad)	(22)	South Africa	0, M2, M4, M6	Plasma (frozen)	Smear microscopy (NR) Liquid culture (MGIT 960) Line probe assay (Hain)
	Kumar (2016)	VEGF-A, VEGF-C, VEGF-R1, VEGF-R2 and VEGF-R3 using the Duoset ELISA Development System (R&D Systems)	(76)	India	0, M6	Platelet-poor plasma (fresh)	Culture (NR)
	Riou (2012)	VEGF – Multiplex bead array (Bulletin# 10014905 Bio-Rad) on a luminometer (Luminex)	(35)	South Africa	0, W2, W4, W8, W12, W26, W52, W78	Plasma (frozen)	Solid culture (LJ) Liquid culture (MGIT) Smear microscopy (FM)
Vitamin B12	Gebremicael (2019)	e601 module for Vitamin B12 on the cobas6000 analyser (Roche Diagnostics Corporation)	(47)	Ethiopia	0, M6	Serum (frozen)	Clinical outcome Smear microscopy (ZN)
White blood cell (WBC) count	Chedid (2020)	Bangladesh: Callatac ES Automated Hematology Analyzer, MEK-7300 (Nihon Kohden) Georgia: XT-20001 (Sysmex) Lebanon: CELL-DYN Ruby (Abbot Madagascar: XT2100i and XN1000 (Sysmex) Paraguay: XT-20001 (Sysmex)	(77)	Bangladesh, Georgia, Lebanon, Madagascar, Paraguay	0, M2, EOT, EOT+2M	Whole blood (NR)	Solid culture (LJ) Liquid culture (MGIT) Smear microscopy (ZN) Smear microscopy (FM)

Table S6: Full list of studies and references included in the quantitative fold-change analyses

	Week 8 fold-change		Weekly fold-change (meta-regression)	
Biomarker	**Author (Year)***	**Ref**	**Author (Year)**	**Ref**
CRP	Almeida (2009)	(20)	Almeida (2009)	(20)
	Ferrian (2017)	(22)	Ferrian (2017)	(22)
	Jayakumar (2015)	(24)	Francisco (2017)	(23)
	Khalil (2020)	(25)	Jayakumar (2015)	(24)
	Miranda (2017)	(27)	Khalil (2020)	(25)
			Miranda (2017)	(27)
			De Moraes (2014)	
IL-6	Chowdhury (2014)	(42)	Chowdhury (2014)	(42)
	Feng (2020)	(51)	Djoba Siawaya (2009)	(34)
	Luo (2018)	(49)	Feng (2020)	(51)
	Mvungi (2019)	(46)	Luo (2018)	(49)
			Mvungi (2019)	(46)
IP-10	Chen (2011)	(40)	Chen (2011)	(40)
	Ferrian (2017)	(22)	Djoba Siawaya (2009)	(34)
	Jayakumar (2015)	(24)	Ferrian (2017)	(22)
	Kim (2018)	(58)	Francisco (2017)	(23)
			Garcia-Basteiro (2017)	(55)
			Hong (2013)	(56)
			Jayakumar (2015)	(24)
			Kim (2018)	(58)
			Riou (2012)	(35)
TNF-α	Chowdhury (2014)	(42)	Chowdhury (2014)	(42)
	Luo (2018)	(49)	Djoba Siawaya (2009)	(34)
	Mvungi (2019)	(46)	Luo (2018)	(49)
	Nie (2020)	(44)	Mvungi (2019)	(46)
			Nie (2020)	(44)
			Zhu (2015)	(54)

* Only papers that included week 8 data were included in this analysis.

Figure S7: Fold-changes plots for individual studies of CRP, IP-10, IL-6 and TNF-α

CRP

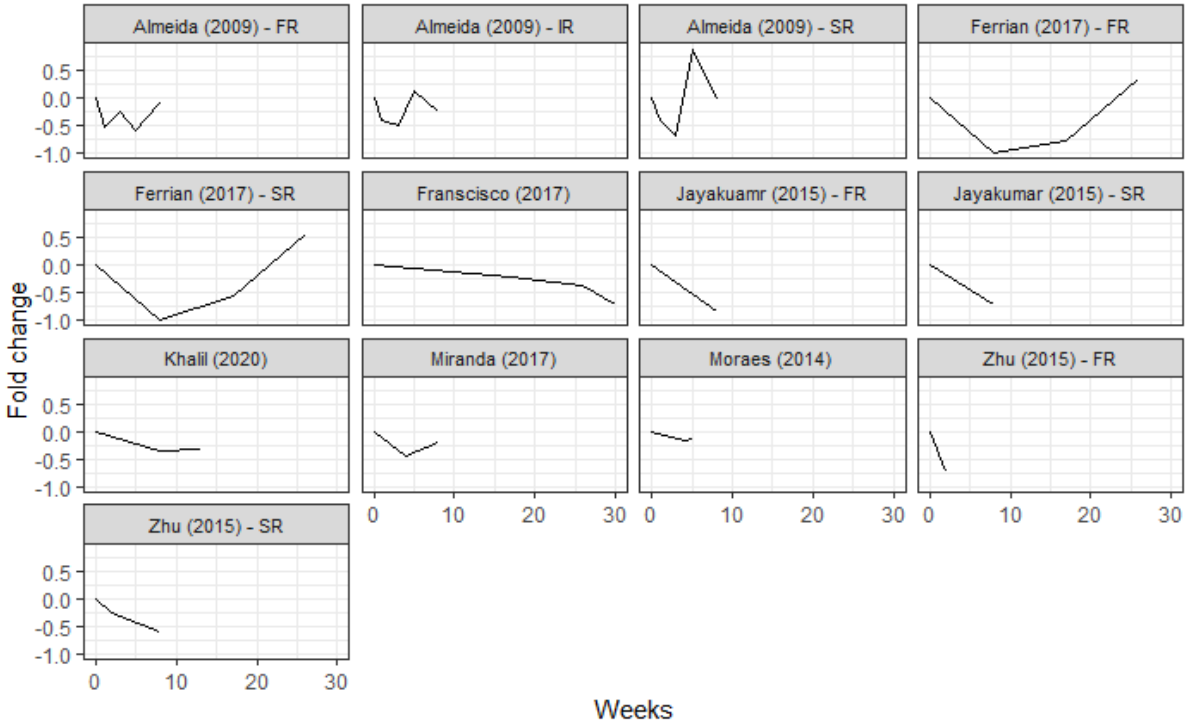

Figure S7.1: CRP fold change among TB patients on treatment across different studies. Fold change is relative to the previous time point reported in the study. FR – fast responders; IR – intermediate responders, SR – slow responders.

IL-6

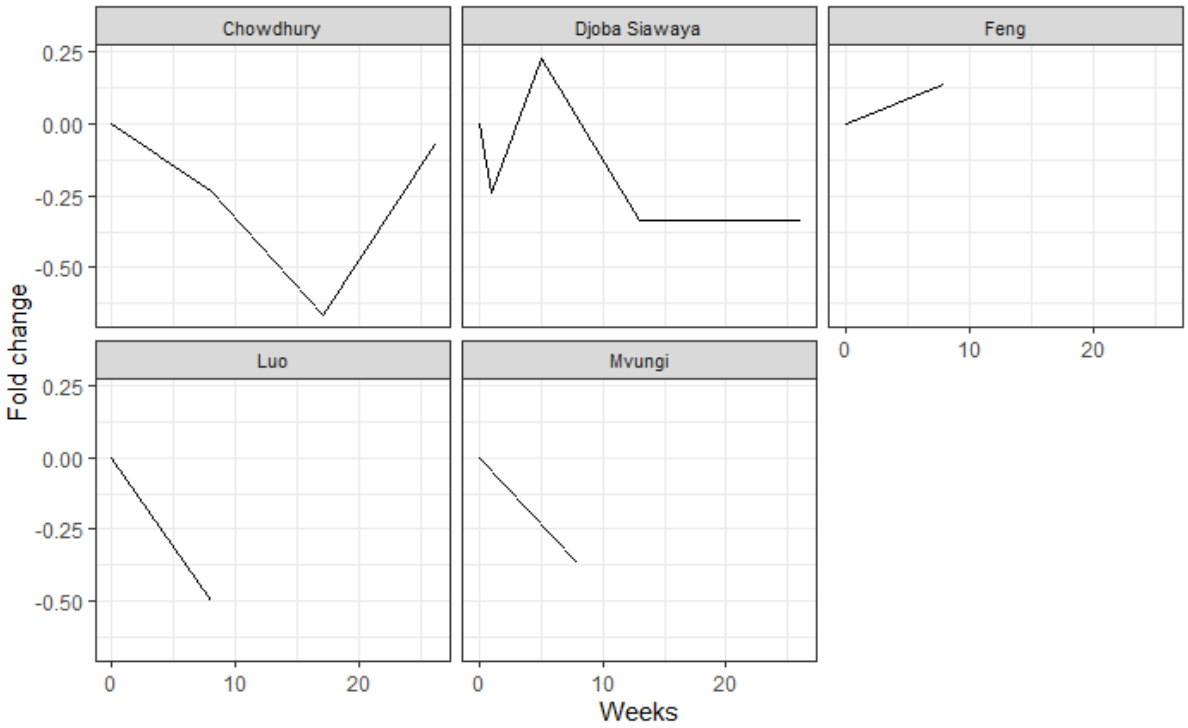

Figure S7.2: IL-6 change among TB patients on treatment across different studies. Fold change is relative to the previous time point reported in the study.

IP-10

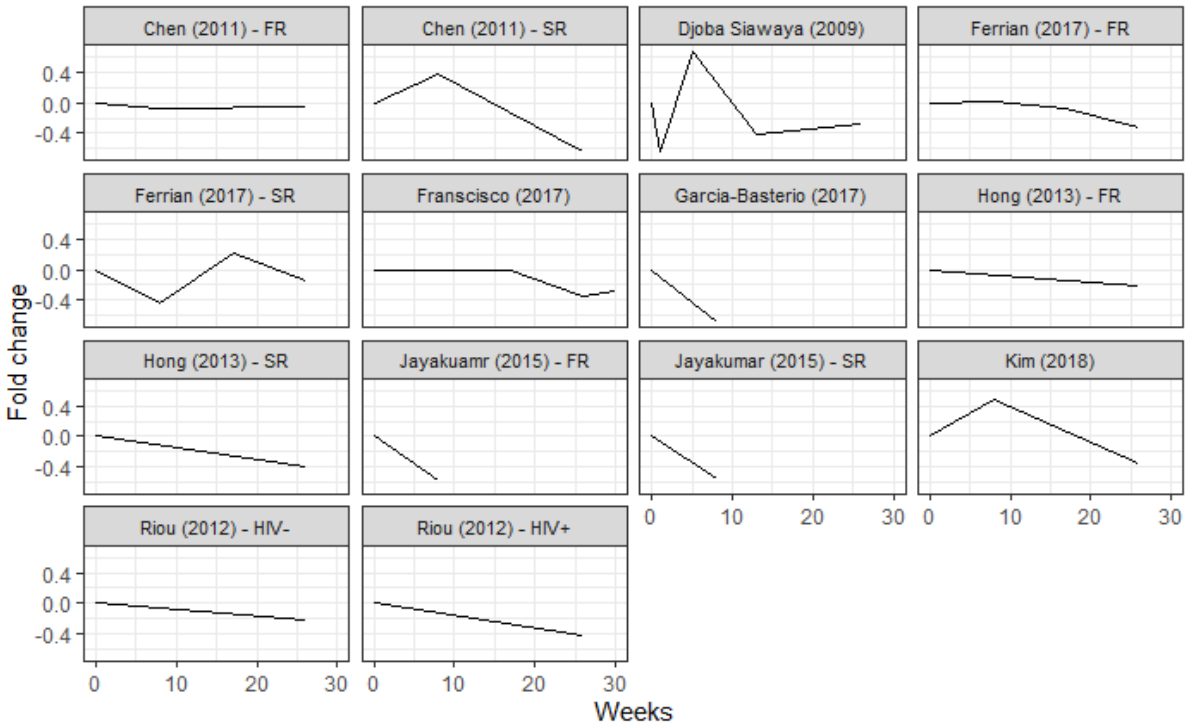

Figure S7.3: IP-10 fold change among TB patients on treatment across different studies. Fold change is relative to the previous time point reported in the study. FR – fast responders; SR – slow responders.

TNF-α

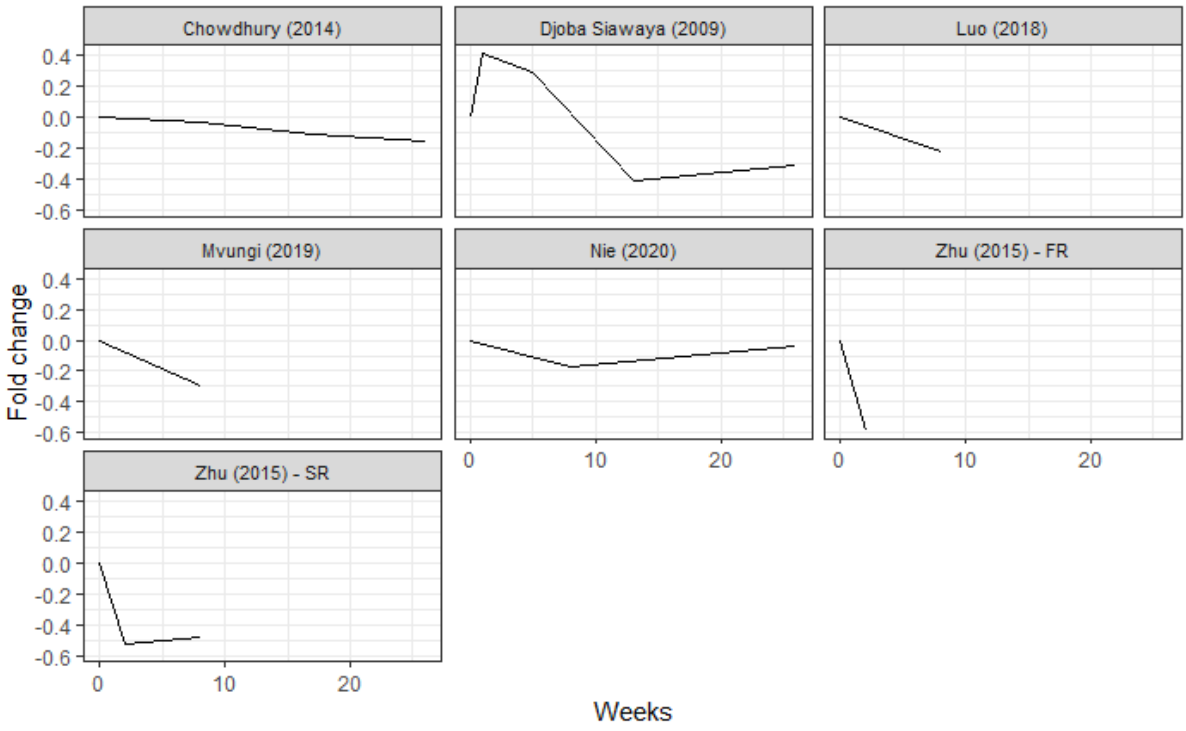

Figure S7.4: TNF-α fold change among TB patients on treatment across different studies. Fold change is relative to the previous time point reported in the study. FR – fast responders; SR – slow responders.

Table S8: Sensitivity analysis for the weekly fold-change meta-regression with varying correlation coefficient (ρ) parameter

	Correlation coefficient (ρ) (95% CI)			
	0	**0.25**	**0.5**	**0.75**
CRP	-0.546 (-0.702, -0.390)	-0.547 (-0.705, -0.389)	-0.539 (-0.702, -0.375)	-0.558 (-0.714, -0.401)
IL-6	-0.317 (-0.595, -0.038)	-0.313 (-0.596, -0.031)	-0.310 (-0.595, -0.025)	-0.306 (-0.592, -0.019)
IP-10	-0.357 (-0.497, -0.218)	-0.361 (-0.495, -0.227)	-0.362 (-0.490, -0.234)	-0.367 (-0.494, -0.241)
TNF-α	-0.119 (-0.171, -0.067)	-0.150 (-0.248, -0.053)	-0.177 (-0.313, -0.040)	-0.149 (-0.227, -0.071)

Table S9: Meta-analysis of outcome is the fold-change from previous recorded timepoint sensitivity analysis for CRP (excluding Ferrian et al.)

Biomarker	Estimate (95% CI)
CRP[1]	-0.539 (-0.702, -0.375)
CRP[2]	-0.453 (-0.594, -0.313)

[1]Including Ferrian et al.
[2]Excluding Ferrian et al.

Table S10: Meta-regression where the outcome is the fold-change from baseline and the predictor is treatment duration.

Biomarker	Estimate (95% CI)
CRP[1]	-0.0001 (-0.0002, 0.0001)
CRP[2]	-0.0518 (-0.0552, -0.0484)
IL-6	-0.0158 (-0.0160, -0.0156)
IP-10	-0.0320 (-0.0354, -0.0285)
TNF-α	-0.0094 (-0.0137, -0.0051)

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